

MTH622 MID TERM SOLVED PAPER BY VURANK
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Question # 6 of 10 | Start time: 03:22:38 PM, 13 June 2019 |

Total Marks: 1

Greens theorem applies only to _____ vector fields

Select the correct option

☐

one dimensional

☐

two dimensional



☐

three dimensional

☐

None of these

Time left: 03:22:38 PM, 13 June 2019

SUBSCRIBE

Scanned by CamScanner

Which of the following theorem convert line integral to surface integral?

Select the correct option

- | | |
|-----------------------|--------------------------------------|
| ☐ | Stoke's theorem only |
| ☐ | Gauss divergence and Stoke's theorem |
| ☐ | Green's theorem only |
| ☐ | Stoke's and Green's theorem |

Click & drag to move this question

SUBSCRIBE

The rate of change of tangential velocity of an object traveling in a circular orbit or path is known as

Select the correct option

☐

Normal acceleration

☐

Tangential acceleration

☐

Rectangular acceleration

☐

None of these

Click to view answer & Mark of this question

Applications of Green's theorem are meant to be in

Select the correct option

☐

One dimensional

☐

Two dimensional

☐

Three dimensional

☐

Four dimensional

Back to Question | Mark as Incomplete

The normal component of acceleration is zero when motion of particle is

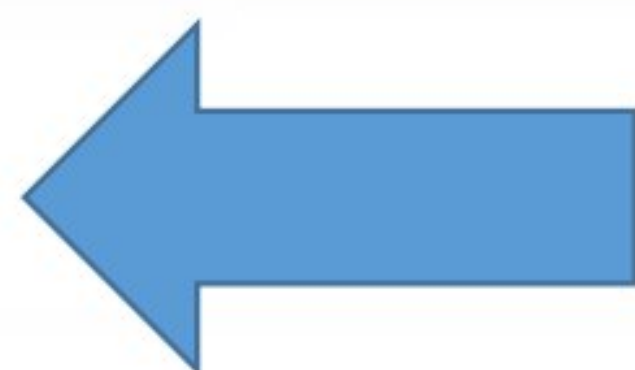
Select the correct option

☐

curvilinear

☐

rotational

☐

rectilinear

☐

translation

Question 13 of 13 | 100% correct

The Stoke's theorem uses which of the following operation?

Select the correct option

☐	Divergence
☐	Gradient
☐	Curl
☐	Laplacian



Time: 02:54:22 PM, 12 June 2019

Greens theorem applies only to _____ vector fields

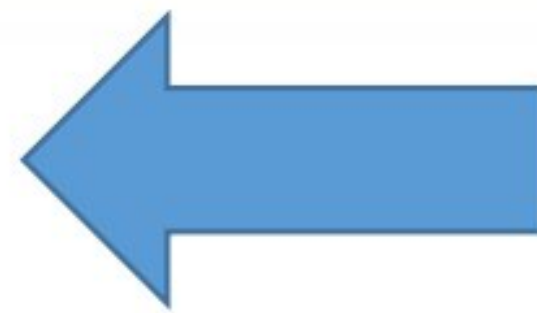
Select the correct option

☐

one dimensional

☒

two dimensional

☐

three dimensional

☐

None of these

Click to see the correct answer

The radial component of velocity for a particle moving in circular path is

Select the correct option



constant



radius itself



variable



zero



Click to Save Answer & Move to Next Question

The Stoke's theorem can be used to find which of the following?

Select the correct option

- | | |
|-----------------------|---|
| ☐ | Surface area enclosed by a function in the given region |
| ☐ | Volume enclosed by a function in the given region |
| ☐ | Linear distance |
| ☐ | Curl of the function |

Click to find correct & Mark as my answer

The Stoke's theorem uses which of the following operation?

Select the correct option

☐	Divergence
☐	Gradient
☐	Curl
☐	Laplacian



11:56:22 PM 13 June 2019

Greens theorem applies only to _____ vector fields

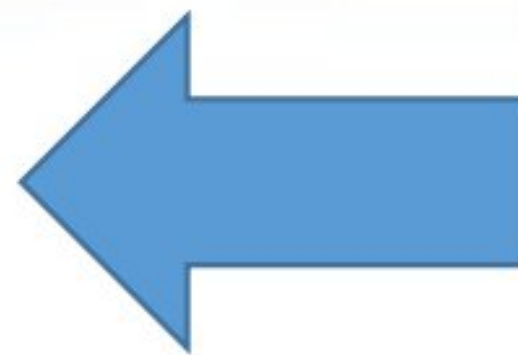
Select the correct option



one dimensional



two dimensional



three dimensional

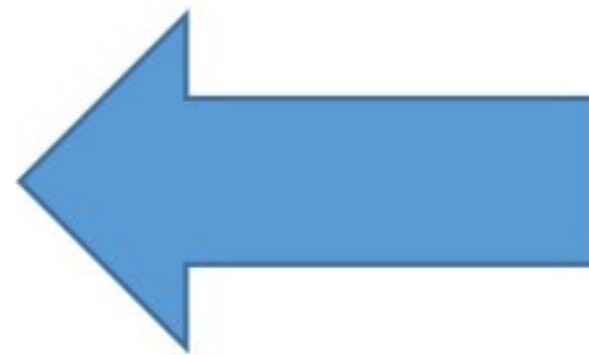


None of these

The point, through which the whole weight of the body acts, irrespective of its position, is known as

Select the correct option

☐	Moment of inertia
☐	Center of gravity
☐	Center of percussion
☐	Center of mass



Time taken to answer this question: 00:00:00

Displacement of a body is a _____ quantity

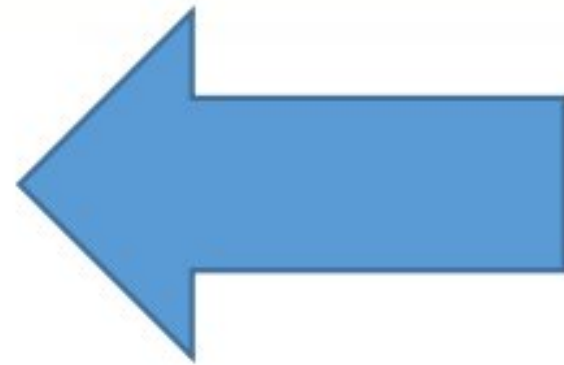
Select the correct option



scalar



vector



scalar and vector



none of these

Click a button below to mark your answer

Which of the following is NOT an application of Green's theorem?

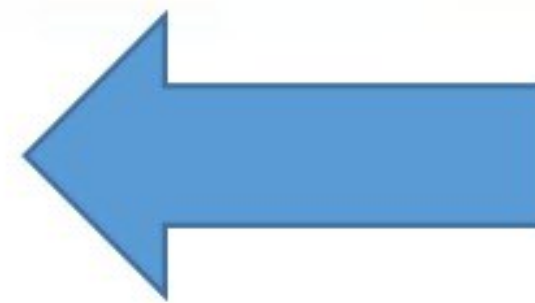
Select the correct option

☐

Centroid of plane figures

☐

Volume of plane figures

☐

Area surveying

☐

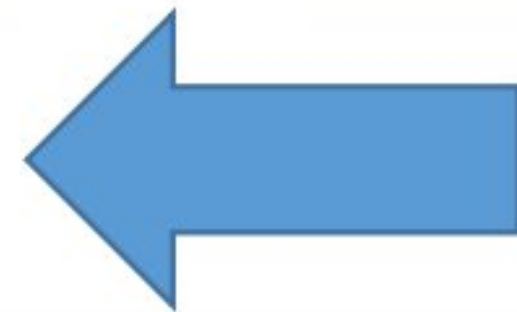
Solving two dimensional flow integrals

Back to Main Menu | View Question

Which of the following is the simplest type of motion along a straight line path.

Select the correct option

- | | |
|-----------------------|--------------------|
| ☐ | plane motion |
| ☐ | rectilinear motion |
| ☐ | curvilinear Motion |
| ☐ | none of these |



Submit Answer

The normal component of acceleration is zero when motion of particle is

Select the correct option



curvilinear



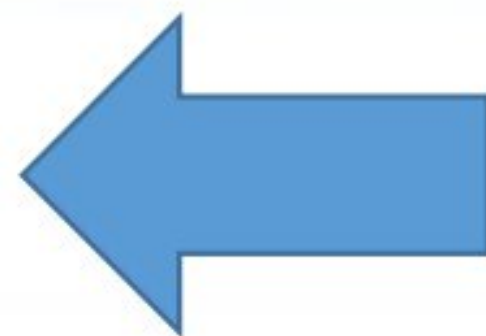
rotational



rectilinear



translation



Back to Question Bank / Mark as Question

The Green's theorem can be related to which of the following theorems mathematically?

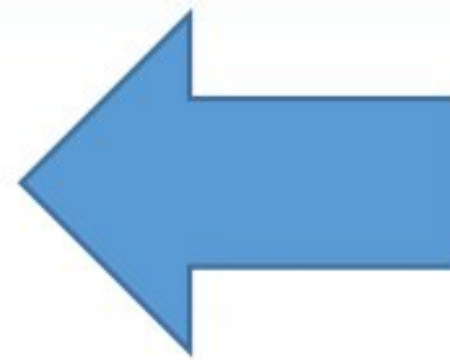
Select the correct option

☐

Gauss divergence theorem

☐

Stoke's theorem

☐

Euler's theorem

☐

Leibnitz's theorem

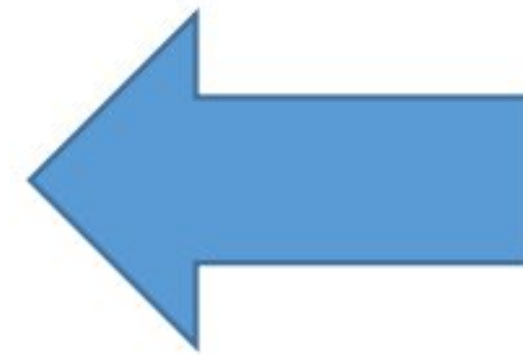
Save & Submit Answer & View Solution

When a particle moves along a straight path, then the particle has

Select the correct option



tangential acceleration only



centripetal acceleration only



both tangential and centripetal acceleration



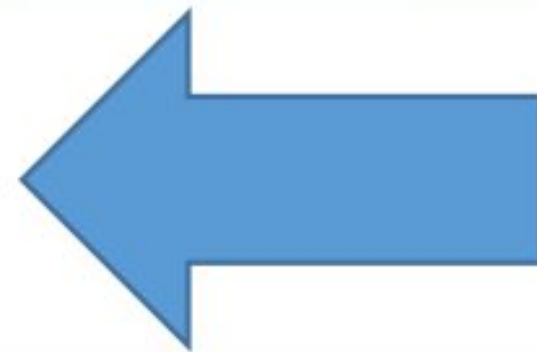
none of these

Back to Question | Marked as Answered

The Stoke's theorem uses which of the following operation?

Select the correct option

☐	Divergence
☐	Gradient
☐	Curl
☐	Laplacian



View Question and Answer | Report Question

The normal component of acceleration is zero when motion of particle is

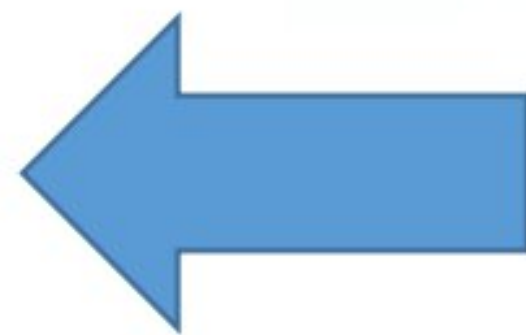
Select the correct option



curvilinear



rotational



rectilinear



translation

The Stoke's theorem can be used to find which of the following?

Select the correct option

- | | |
|----------------------------------|---|
| ☐ | Surface area enclosed by a function in the given region |
| ☒ | Volume enclosed by a function in the given region |
| ☐ | Linear distance |
| ☐ | Curl of the function |



Click to Show Answer & Move to Next Question

Which of the following is NOT an application of Green's theorem?

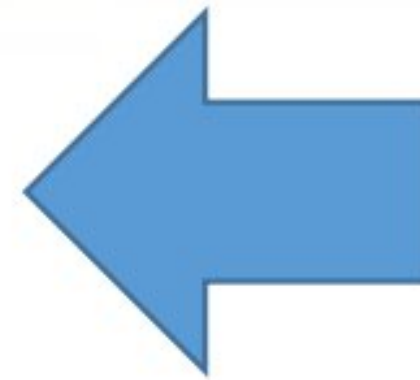
Select the correct option



Centroid of plane figures



Volume of plane figures



Area surveying



Solving two dimensional flow integrals

The Green's theorem can be related to which of the following theorems mathematically?

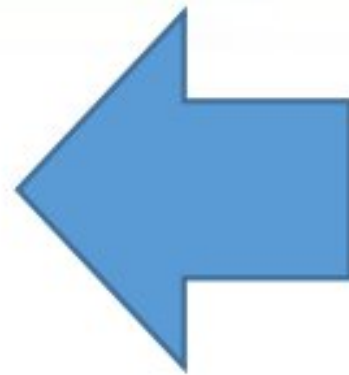
Select the correct option

☐

Gauss divergence theorem

☐

Stoke's theorem

☐

Euler's theorem

☐

Leibnitz's theorem

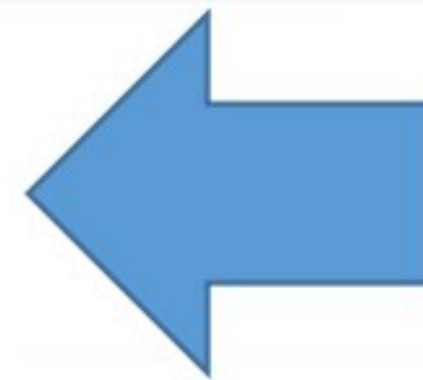
Time: 03:54:56 PM, 13 June 2019

Which of the following theorem convert line integral to surface integral?

Select the correct option

☐

Stoke's theorem only

☐

Gauss divergence and Stoke's theorem

☐

Green's theorem only

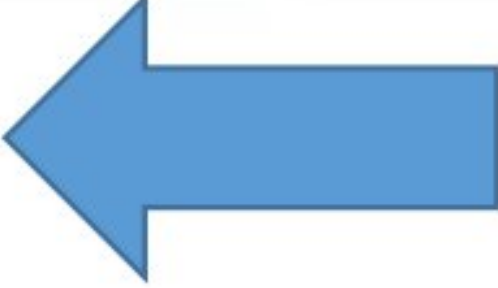
☐

Stoke's and Green's theorem

Submit Answer & Mark as Correct

When a particle moves along a straight path, then the particle has

Select the correct option

- | | | |
|-----------------------|--|---|
| ☐ | tangential acceleration only |  |
| ☐ | centripetal acceleration only | |
| ☐ | both tangential and centripetal acceleration | |
| ☐ | none of these | |

Submit your answer & Go to Next Question

Applications of Green's theorem are meant to be in

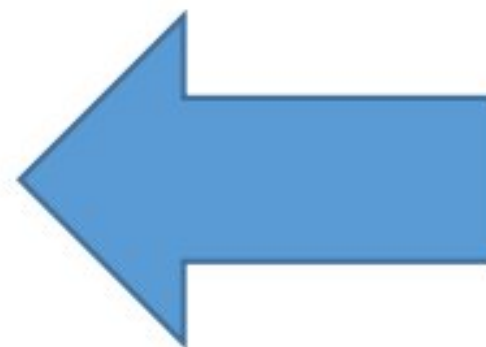
Select the correct option



One dimensional



Two dimensional



Three dimensional

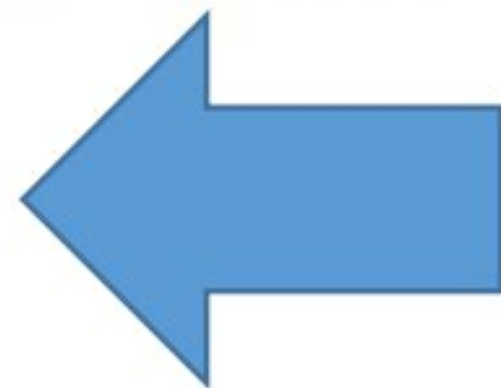


Four dimensional

Which of the following is the simplest type of motion along a straight line path.

Select the correct option

- | | |
|-----------------------|--------------------|
| ☐ | plane motion |
| ☐ | rectilinear motion |
| ☐ | curvilinear Motion |
| ☐ | none of these |



Submit your answer

The Stoke's theorem can be used to find which of the following?

Select the correct option

☐

Surface area enclosed by a function in the given region

☐

Volume enclosed by a function in the given region

☐

Linear distance

☐

Curl of the function

Navigation buttons: Previous, Next, etc.

Question 9 of 10 (Start time: 02:46:17 PM, 13 June 2019)

Both double and triple integrals can be used to calculate

Select the correct option

☐	area under the curve
☐	length of curve
☐	volume of the given object
☐	None of these

Submit Answer

The normal component of acceleration is zero when motion of particle is

Select the correct option

☐	curvilinear
☐	rotational
☐	rectilinear
☐	translation

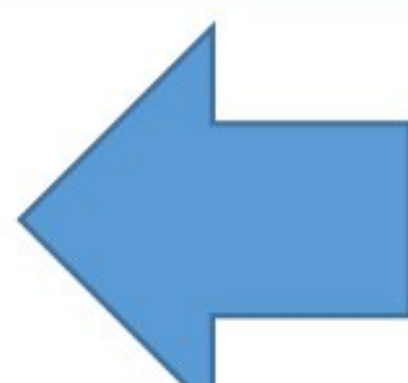


Time is now 03:28:11 PM. 13 June 2019

The radial component of velocity for a particle moving in circular path is

Select the correct option

☒	constant
☐	radius itself
☐	variable
☐	zero



Stop to Review Answer & Move to Next Question

Which of the following is NOT an application of Green's theorem?

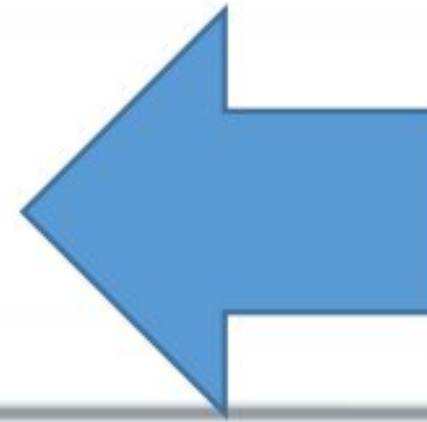
Select the correct option



Centroid of plane figures



Volume of plane figures



Area surveying



Solving two dimensional flow integrals

Displacement of a body is a _____ quantity

Select the correct option



scalar



vector



scalar and vector



none of these

Back to Question Bank & More to Your Library

Greens theorem applies only to _____ vector fields

Select the correct option



one dimensional



two dimensional



three dimensional



None of these



Submit Answer

The radial component of velocity for a particle moving in circular path is

Select the correct option



constant



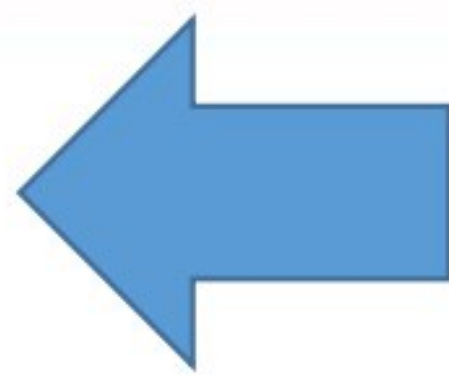
radius itself



variable



zero

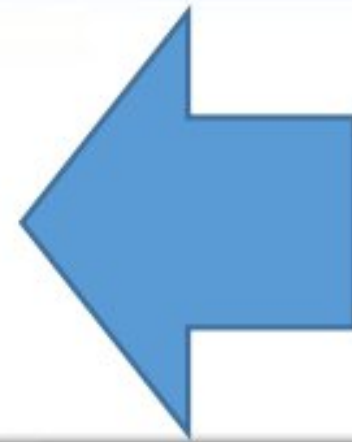


Go to the question & answer list

The point, through which the whole weight of the body acts, irrespective of its position, is known as

Select the correct option

- | | |
|-----------------------|----------------------|
| ☐ | Moment of inertia |
| ☐ | Center of gravity |
| ☐ | Center of percussion |
| ☐ | Center of mass |



The rate of change of tangential velocity of an object traveling in a circular orbit or path is known as

Select the correct option



Normal acceleration



Tangential acceleration



Rectangular acceleration



None of these

Using CamScanner

According to Stokes' Theorem if S is an open, non-orientable surface bounded by a simple closed curve C , and

$$\vec{A}$$

has continuous first partial derivatives, then

Select the correct option

Submit Answer

☐	$\iint_S (\nabla \cdot \vec{A}) dS$
☐	$\iint_S (\nabla \times \vec{A}) dS$
☐	$\iint_S (\nabla \times \vec{A}) \cdot \vec{n} dS$
☐	$\iint_S (\nabla \times \vec{A}) \cdot \vec{A} dS$

Submit Answer

Question # 7 of 10 (Start time: 09:04:31 PM, 12 June 2019)

Which of the following is NOT an application of Green's theorem?

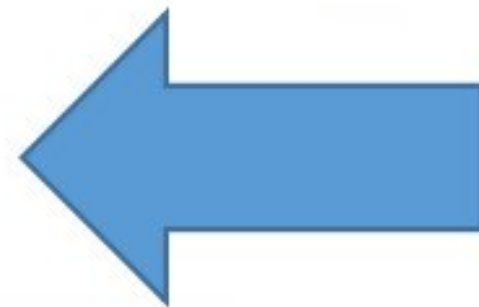
Select the correct option

☐	Centroid of plane figures
☒	Volume of plane figures
☐	Area surveying

The Green's theorem can be related to which of the following theorems mathematically?

Select the correct option

- | | |
|-----------------------|--------------------------|
| ☐ | Gauss divergence theorem |
| ☐ | Stoke's theorem |
| ☐ | Euler's theorem |
| ☐ | Leibnitz's theorem |



View Question History | View Question Details

Question 00 of 10 | Time taken: 00:00:15 PM, 17 June 2021

A fluid of density $\rho(x, y, z, t)$ moves with velocity $\mathbf{v}(x, y, z, t)$. If there are no sources or sinks then

Select the correct option



$$\nabla \cdot (\rho \mathbf{v}) + \frac{\partial \rho}{\partial t} = 0$$



$$\frac{\partial \rho}{\partial t} = 0$$



$$\nabla \cdot (\rho \mathbf{v}) = 0$$



$$\nabla \cdot (\rho \mathbf{v}) + \frac{\partial \rho}{\partial t} = 0$$

Classical mechanics deals with the

Select the correct option



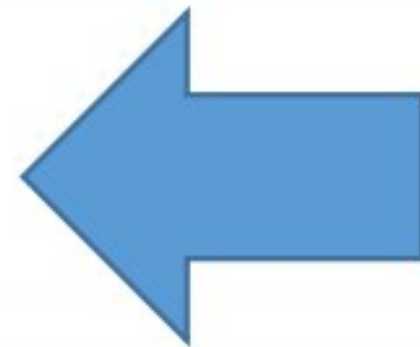
Statics objects only



microscopic objects only



macroscopic objects only



None of these

MTH622 Quiz #1 MTH622

Question # 3 of 10 (Start time: 09:02:43 PM, 12 June 2019)

Displacement of a body is a _____ quantity

Select the correct option

☐	scalar
☒	vector

Question # 5 of 10 (Start time: 09.03.12 PM, 12 June 2019)

Both double and triple integrals can be used to calculate

▶ Select the correct option

☐	area under the curve
☐	length of curve
☐	volume of the given object



$$y = (4-x)^{1/2}$$

$$\frac{dy}{dx} = \frac{1}{2} (4-x)^{\frac{1}{2}-1} \cdot \frac{d}{dx} (4-x)$$

$$= \frac{1}{2} (4-x)^{-1/2} (-1)$$

$$= \frac{-1}{2(4-x)^{1/2}}$$

$$= \frac{-1}{2\sqrt{4-x}}$$

$$\ln y = \frac{1}{2} \ln(4-x)$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{2} \left(\frac{-1}{4-x} \right) + \ln(4-x) \cdot \frac{1}{2}$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{-1}{2(4-x)} + \frac{\ln(4-x)}{2}$$

$$= \frac{-1}{2(4-x)} + \frac{\ln(4-x)}{2}$$

$$= \frac{-1 + \ln(4-x)}{2(4-x)}$$

$$\frac{dy}{dx} = y \left(\frac{-1 + \ln(4-x)}{2(4-x)} \right)$$

$$\frac{dy}{y} = \frac{1}{2} \left(\frac{-1 + \ln(4-x)}{4-x} \right) dx$$

$$\frac{1}{2} \left(\frac{-1 + \ln(4-x)}{4-x} \right) dx$$

WTH427 Quiz1 WTH427

Question 8 10 of 10 ([View answer](#) 05:05:13 PM, 12 June 2019)

What moment of inertia of a uniform thin rod of mass M and length l about its end point and perpendicular to its length is

Select the correct option

- | | |
|----------------------------------|---------------------|
| ☐ | $\frac{1}{12} Ml^2$ |
| ☐ | $\frac{1}{3} Ml^2$ |
| ☐ | $\frac{1}{2} Ml^2$ |
| ☒ | $\frac{1}{3} Ml^2$ |

Math 622 25/6/18

* State Green's Theorem

AMRAB M-SC (MATHS)

GOLD MEDALIST

* Normalize $3x + 2y + z = 4$

math2IT3@gmail.com

* If $A = (z^2 - 2xy)\mathbf{i} + 2xz\mathbf{j}$ then find curl of A

* If mass and momentum given that Force field find kinetic energy

* If A is irrotational then find divergence theorem

Mth622

Mcqs easy thy tough ni thy

- 1. Divergence theorem ka question**
- 2. Stocks theorem ka question tha**
- 3. Irrotational proof tha 1 vector ko**
- 4. Newton 2nd law of motion**
Baql 2 question me proof thy 2 choty choty

Mth622

Objectives conceptual

- 1.newton second law of motion**
- 2.del. $A=0$ AND $\text{del}^*A=0$...**
Justify
- 2.A was given.. Find del. A**
- 3.FOrce find mas and acceleration givn thi.**
- 4.line integral find**
- 5.Stoks law**

Mth622

Surface integrl k 2 que
lk directional derivative



4. Newton 2nd law of motion
Baqi 2 question me proof thy 2
choty choty

Mth622

Objectives conceptual

1.newton second law of motion

2.del. $A=0$ AND $\text{del}^*A=0$...

Justify

2.A was given.. Find del. A

3.FOrce find m and a and
acceleration givn thi.

4.line integral find

5.Stoks law

Mth622

Surface integrals 2 que

1 directional derivative

1 koi normal surface wala tha

Volume integrals long que tha

1 green theorem sa que tha

Write statemt of green theorem



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(KEEP REMEMBER IN YOUR PRAYERS)